

Project Proposal

Title: Precision Biological Illustration Generation via Multi-Agent Collaboration and Knowledge Base

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I. INTRODUCTION

Text-to-image diffusion models have shown remarkable synthesis capabilities, such as Gemini, ChatGPT, but often fail in domain-specific tasks requiring high structural integrity. In biological sciences, "hallucinations", such as misplacing a protein receptor or mislabeling a bone name, let AI-generated images educationally invalid. This project addresses these limitations by introducing BioGuard-Diffusion, a "Reasoning-before-Generation" pipeline. Our approach builds upon the importance of semantic parsing in scientific images, questions and annotations established by the AI2D dataset and the potential of Retrieval-Augmented Generation to improve synthesis. By leveraging Multi-Agent Collaboration frameworks, we decompose each steps into specialized agents, significantly reducing hallucinations compared to single-prompt inference and ensuring the anatomical accuracy of generated biological illustrations.

II. PROPOSED METHODOLOGY

The BioGuard-Diffusion system is divided into two subsequent reasoning and generation stages:

A. Reasoning Layer (Multi-Agent System)

We plan to use LangChain as our Multi-Agent framework to create different agents.

- The Biologist Agent: Put users' queries into RAG system to retrieve specific biological requirements and knowledge (e.g., "A mitochondria must have inner folds/cristae").
- The Spatial Architect: Uses logic to create a structural layout prompt for diffusion model to follow (e.g., "Human's hand has five finger").
- The Verifier Agent: A Vision-Language Model (VLM) that cross-checks the draft against scientific facts before final output are sent to user.

B. Generation Layer

This layer is designed to use text-to-image diffusion model to generate image. Since the cutting-edge model, like Gemini, ChatGPT, will be very expensive. We might choose to find some open-source diffusion model on huggingface like stable-diffusion-v1-5-GGUF or Z-Image as our diffusion model. In the final stage, we can also test this agent flow with cutting-edge model to see the outcome. This layer should receive the spatial constraints and structural skeletons in prompt from agents.

III. EXPERIMENTS AND EVALUATION

To validate the effectiveness of BioGuard-Diffusion, we will compare three configurations: (1) One-shot Diffusion Model generate, (2) RAG + Diffusion Model, and (3) Full Multi-Agent BioGuard-Diffusion to prove the necessity of multi-agent workflow. We have not decided the whole evaluation matrix. We plan to use following matrix, Scientific Accuracy Score (SAS), CLIP-Score and Bio-CLIP.